

A Communication-Efficient Compression Method for FedDyn Federated Learning Models

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Abstract—In the practical applications of federated learning, client data often exhibits highly non-independent and identically distributed characteristics, leading to unstable convergence during model training. FedDyn effectively mitigates client drift problem by introducing dynamic regularization terms, but it requires the complete transmission of model parameters in each round of transmission, resulting in significant communication cost and limiting its application in communication-constrained scenarios. To address the aforementioned issues, this paper proposes a communication-efficient method for compressing FedDyn federated learning models. This method introduces parameter importance evolution strategy without changing the FedDyn dynamic regularization optimization mechanism, and only communicates key parameters in global model updates, thereby reducing the overall communication cost. The experimental results show that on Non-IID datasets such as MNIST, FashionMNIST, and Cifar10, the proposed method can decrease the overall communication overhead by about 70%-85% while maintaining stable convergence performance, without a decrease of more than 5% in the final model accuracy. This verifies the effective balance between model performance and communication efficiency.

Keywords—Federated learning, FedDyn, Model compression, Communication efficiency, Non-IID data

I. INTRODUCTION

With the rapid development of mobile terminals and edge computing devices, data gradually displays characteristics of distributed storage and privacy sensitivity. Federated learning provides an effective solution for distributed learning in scenarios of privacy protection by enabling collaborative model training without sharing original data [1]. However, in practical applications, the data held by each client often exhibits non-independent and identically distributed characteristics, making federated learning models prone to slow convergence and unstable performance during training [2].

To overcome the challenges posed by Non-IID data, researchers proposed several improvement methods. Among them, FedDyn builds client historical update information explicitly by introducing dynamic regularization terms in the local objective function, mitigating the problems of client drift caused by data distributed difference effectively. Existing research has proved that compared to methods such as FedAvg, FedProx and MOON, FedDyn is equipped with faster convergence speed, higher final accuracy and more stable training process in various Non-IID scenarios [3-5].

However, while FedDyn achieved performance advantages, it also pays higher communication costs. Specifically, FedDyn needs to transmit the complete model

parameters in each round of communication, and the amount of communication information increases proportionally with model scale and the number of communication rounds [6]. When the model is large or communication resources are limited, high communication costs will become a key factor restricting the practical deployment of FedDyn.

Therefore, this paper studies the model compression problem under the FedDyn framework from the perspective of communication efficiency, proposing a communication compression method that matches the FedDyn dynamic regularization mechanism. This method reduces the overall communication costs of the system without destroying the original optimization structure and Non-IID advantage of FedDyn, providing a more feasible solution for the application of FedDyn in practical federated learning systems.

II. RELATIVE WORKS

A. Federated Learning and Non-IID Problems

FedAvg, a classic method of federated learning, achieves global model update by performing a weighted average of client model parameters on the server side. Conversely, under the Non-IID data scenarios, FedAvg is always influenced by client drift problems, which leads to unstable model convergence [7,8]. To alleviate this problem, researchers proposed solutions such as FedProx and MOON, which introduce regularization terms or contrastive learning mechanisms to improve model performance. FedProx restricts the drift between local model and global model by introducing proximal regularization in the local optimization objective; MOON, on the other hand, uses a contrastive learning mechanism to intensify the representation consistency between different client models. These methods alleviate performance degradation problems in Non-IID scenarios to some extent, but they still have certain limitations in complex heterogeneous environments, and still have room for further optimization in the balance of communication costs and model performance.

B. FedDyn Method

FedDyn is a federated learning optimization method designed for Non-IID scenarios, which explicitly models historical update information on the client side by introducing dynamic regularization terms in local objective functions, effectively mitigating client drift problems from theoretical and experimental perspectives.

Equation (1) defines the local optimization objective of client k in FedDyn in round t . By introducing dynamic regularization terms in the local loss function and using historical gradient information to restrict model updates, it

mitigates client drift problems caused by non-independent and identically distributed data. Equation (2) gives the first-order optimality condition of the local optimization problem, and reveals the constraint effect of dynamic regularization terms on local update at the gradient level, making the model update consistent and improving overall stable convergence.

$$\theta_k^t = \operatorname{argmin} \left[L_k(\theta) - \langle \nabla L_k(\theta_k^{t-1}), \theta \rangle + \frac{\alpha}{2} \|\theta - \theta^{t-1}\|^2 \right] \quad (1)$$

$$\dot{\theta}_k^t = \nabla L_k(\theta_k^t) - \nabla L_k(\theta_k^{t-1}) + \alpha(\theta_k^t - \theta^{t-1}) \quad (2)$$

Compared to FedAvg and FedProx, FedDyn shows more stable convergence behaviour and higher model accuracy in several non-independent and identically distributed scenarios.

C. Communication Compression Techniques in Federated Learning

In order to decrease the communication overhead in federated learning systems, several communication compression techniques have been proposed, including parameter pruning, quantization and sparsification. This type of method originated from research on stochastic gradient descent for communication, which reduces communication costs by decreasing gradient and transmission accuracy or frequency [9,10]. This method can reduce communication costs to some extent by transmitting only a portion of model parameters or applying low-precision representation.

However, most existing communication compression methods are based on simple parameters amplitudes and fixed rules for compression, which often seriously influence convergence performance and prediction accuracy under the scenarios of highly heterogeneous data distribution. The characteristics of different compression methods are shown in Table I. In the table, “✓” indicates that the method is equipped with good adaptability in certain dimensions, whereas “×” indicates that the characteristic has not been explicitly considered in the existing methods. When introducing complex optimization mechanisms such as dynamic regularization, directly applying conventional communication compression strategies may destroy original optimization structures, which leads to the significant decrease in model accuracy.

Therefore, how to apply communication compression mechanisms effectively into complex federated optimization algorithms while ensuring model accuracy and stable convergence remains a problem worthy of further research.

TABLE I. COMPARISON OF CHARACTERISTICS OF DIFFERENT FEDERATED LEARNING COMMUNICATION COMPRESSION METHODS IN NON-IID SCENARIOS

Methods	FedDyn-aware	Parameter importance	Extra parameter	Performance influence
Top-k	×	×	K	Intermediate
Quantization	×	×	Bit	Intermediate
Periodic Upload	×	×	T	Low
FedDyn-FPI(ours)	✓	✓	ρ	High

III. RESEARCH METHODS

A. FedDyn-aware Parameter Importance

Starting from the optimized structure of FedDyn, this paper establishes a parameter importance modeling method

that matches its dynamic regularization mechanism, providing theoretical basis for subsequent communication compression strategies.

During the local training period on the client side, this paper completely follows the original optimization framework of FedDyn and does not introduce any compression operations into the local training process. The client uses cross-entropy loss function for model training, and retains the dynamic regularization terms in FedDyn to alleviate client drift problems resulting from non-independent and identically distributed data, thereby ensuring the model can learn sufficient data information locally.

Model compression is performed only during the communication phase. After each round of local training, the client filters model parameters based on parameter importance assessment results and preset communication proportions, uploading only the parameters that contribute significantly to the global model update, while the remaining parameters remain unchanged locally.

To characterize the client's update behaviour relative to the global model during the local training phase, this paper first defines the local model update amount on the client site as shown in formula(3):

$$\Delta\theta_k^t = \theta_k^t - \theta^{t-1} \quad (3)$$

where θ_k^t represents the model parameters obtained by client k after the t-th round of local training, θ^{t-1} represents the global model parameters of the previous round, $\Delta\theta_k^t$ represents the actual update amount of the client relative to the global model in the current communication round. This update amount is also the basic object that needs to be uploaded to the server in subsequent communication phases.

Under the framework of FedDyn, the client's local optimized objective is established by introducing dynamic regularization terms into the loss function. By finding the first-order optimality condition for its local optimization problems and combining it with the definition of local update amount in equation (3), the constraint relationship of FedDyn can be expressed as the inner product form of equation (4):

$$\langle \nabla L_k(\theta_k^t) - \nabla L_k(\theta_k^{t-1}), \Delta\theta_k^t \rangle + \alpha \|\Delta\theta_k^t\|^2 \quad (4)$$

where $L_k(\theta_k^t)$ represents the local loss function of client k, $\nabla L_k(\theta_k^t)$ and $\nabla L_k(\theta_k^{t-1})$ represents the gradient information of the local model in the current round and the previous round respectively, and α is the regularization coefficient.

Based on the relationship between parameter update and gradient change revealed by equation (4), this paper defines the FedDyn-aware parameter importance index (FPI) as shown in equation (5) from the parameter dimension:

$$I_{k,i}^t = |(\nabla L_k(\theta_k^t)_i - \nabla L_k(\theta_k^{t-1})_i) \cdot \Delta\theta_{k,i}^t| \quad (5)$$

where $I_{k,i}^t$ represents the importance measurement of the i-th model parameter of client k in round t, $\nabla L_k(\theta_k^t)_i - \nabla L_k(\theta_k^{t-1})_i$ represents the gradient change of this parameter in the training process of two adjacent rounds, and $\Delta\theta_{k,i}^t$ represents the actual update amount of this parameter in the current round.

After obtaining the importance measurement of each parameter, the client sorts the parameters' importance based on the preset communication compression ratio $\rho \in (0,1]$, and chooses the top ρ parameters to participate in communication. Specifically, the set of selected parameter indices is defined as formula (6):

$$S_k^t = \text{Top}\rho(I_{k,i}^t) \quad (6)$$

Subsequently, the client only uploads the updated amount of the parameters in the set S_k^t , while setting the remaining parameters to zero in the current round of communication. The compressed model update definition is shown in formula (7):

$$\widetilde{\Delta\theta}_{k,i}^t = \begin{cases} \Delta\theta_{k,i}^t, & i \in S_k^t \\ 0, & i \notin S_k^t \end{cases} \quad (7)$$

After the communication phase ends, the server performs weighted aggregation on the compressed updates uploaded by each client to obtain the global model parameters for the next round. The update method is as shown in formula (8):

$$\theta^{t+1} = \theta^t + \sum_{k=1}^K \frac{n_k}{n} \widetilde{\Delta\theta}_{k,i}^t \quad (8)$$

In summary, this paper proposed a FedDyn-aware parameter compression strategy, which reduces the communication cost during model training process effectively by guiding communication process through parameter importance while maintaining the original optimization characteristics of FedDyn.

B. The Complete Communication Compression Process of FedDyn-aware

Based on the FedDyn-aware parameter importance index defined in Section 3.1, this paper designs the compression process of the communication stage under the FedDyn framework, and its overall structure is shown in Figure 1. At the beginning of each round of communication, the server broadcasts the previous round of global model parameters θ^{t-1} to the clients participating in the training. After receiving the global model, the client performs local training according to the FedDyn original dynamic regularization objective function without introducing any compression operations. After training, client k obtains local model parameters θ_k^t , and calculates the local update amount according to Equation (3).

Subsequently, the client evaluates the importance of each model parameter according to the parameter importance index $I_{k,i}^t$ defined in Equation (5) in Section 3.1. This metric takes into account both the parameter update magnitude and the gradient change information between adjacent rounds, and is used to characterize the contribution of different parameters in the global optimization process. After obtaining the importance metrics of all parameters, the client sorts the parameter importance in descending order based on preset communication compression ratio $\rho \in (0,1]$, and chooses the top ρ parameters to form an index set (6).

After determining the set of key parameters, the server only uploads the updated amount of parameters in set S_k^t , while the remaining parameters are set to zero in the current round of communication, thereby constructing the compressed local update vector (Equation (7)).

After the communication phase ends, the server receives all compression updates uploaded by clients, performs weighted aggregation according to the number of samples, and updates global model parameters (Equation (8)).

In summary, the proposed FedDyn-aware communication compression method only transmits the parameters that contribute more to global model update without destroying the original dynamic regularization optimization structure of FedDyn, thereby keeping the balance of communication efficiency and model performance in Non-IID scenarios.

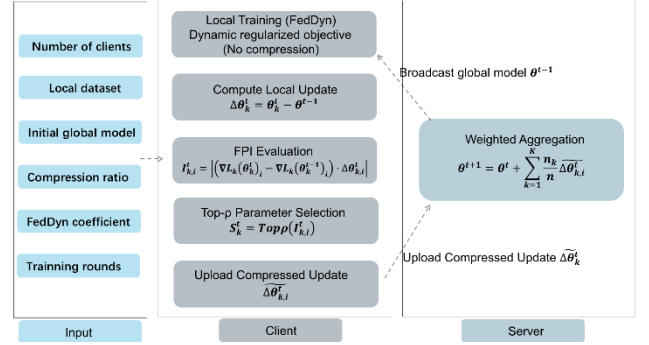


Fig. 1. Diagram of an efficient federated learning framework for communication

IV. EXPERIMENTAL RESULTS AND ANALYSIS

This paper sets the experimental environment and training hyperparameters uniformly, ensuring that all the experiments are performed under the same model structure and data partitioning conditions.

In terms of model structure, this paper uses convolutional neural networks (CNN) as the basic model and conducts experiments on the MNIST, FashionMNIST and Cifar10 datasets. For training parameter settings, local optimization algorithms use the SGD optimizer with a learning rate of 0.01. The number of local training epochs is set to 5, and the batch size is 64. The number of global communication rounds is set to 200. The dynamic regularization coefficient α is set to 0.001 in the FedDyn algorithm.

In the communication compression experiment, communication ratio ρ was set to 1.0, 0.5, 0.2, 0.1, respectively, to analyze the influence of different compression intensities on model performance and communication cost. $\rho = 1.0$ indicates no compression.

A. Performance Comparison of Different Federated Learning Methods

To verify the performance advantage of FedDyn in Non-IID scenarios, this paper first conducted an accuracy comparison experiment on the MNIST, FashionMNIST and Cifar10 datasets, comparing the accuracy of FedDyn, FedAvg, FedProx and MOON methods. The accuracy curves of each algorithm on different datasets are shown in Figure 2, Figure 3, and Figure 4, and the final summary results of the test accuracy are shown in Table II. The results indicate that, with the same number of training rounds, FedDyn exhibits faster convergence speed and higher final model accuracy on all three datasets.

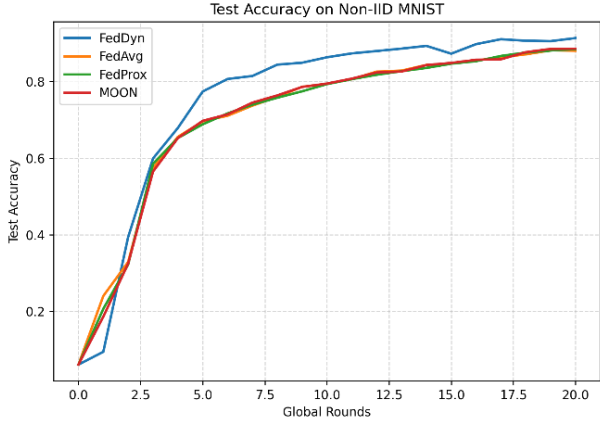


Fig. 2. Accuracy of four algorithms on the MNIST dataset

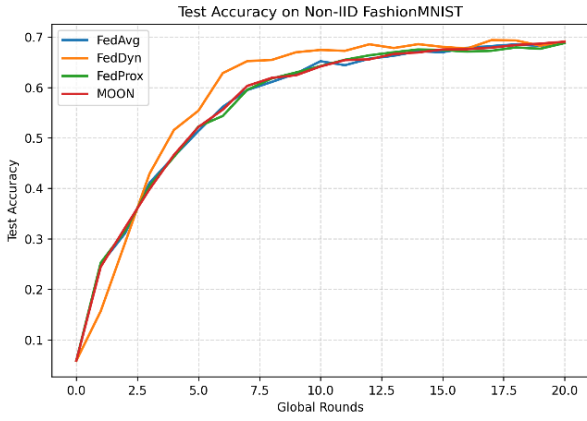


Fig. 3. Accuracy of four algorithms on the FashionMNIST dataset

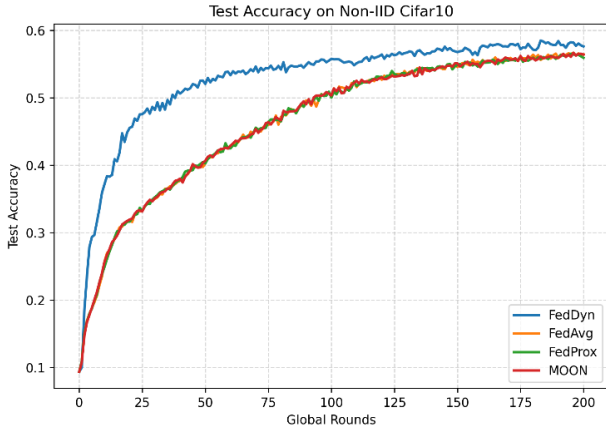


Fig. 4. Accuracy of four algorithms on the Cifar10 dataset

TABLE II. FINAL TEST ACCURACY (%) OF DIFFERENT ALGORITHMS ON THREE DATASETS UNDER NON-INDEPENDENT AND IDENTICALLY DISTRIBUTED (NON-IID) CONDITIONS

Methods	MNIST(%)	FashionMNIST(%)	Cifar10(%)
FedAvg	88.08	68.22	56.36
FedProx	88.04	67.66	56.12
MOON	88.47	68.68	56.50
FedDyn	90.50	69.27	57.75

B. Model Performance Analysis under Different Communication Ratios

Based on verifying the original performance advantages of FedDyn, this paper further analyzes the impact of introducing communication compression on model performance under the different communication ratio settings. In the experiment, the

parameter proportion uploaded from client to server is controlled by adjusting the communication ratio ρ , thereby keeping balance between model performance and communication cost.

The experimental results are shown in Figure 5, Figure 6 and Figure 7. It can be noticed that when the communication ratio is set in a reasonable range, the final accuracy of FedDyn after communication compression only shows a slight decrease and the overall convergence trend is basically consistent with that of the uncompressed FedDyn. This shows that not all model parameters have the same importance to global model update in every round of communication. By reasonably selecting the parameters for communication, FedDyn can significantly reduce the amount of communication data with minimal performance loss.

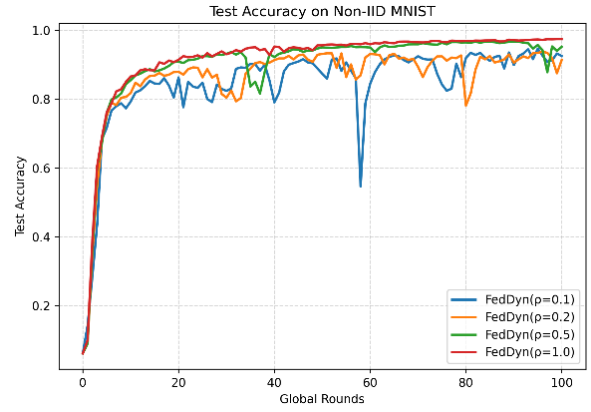


Fig. 5. The test accuracy of FedDyn on Non-IID MNIST dataset under different communication ratio

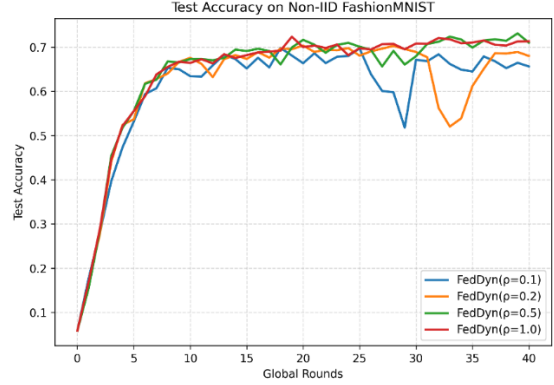


Fig. 6. The test accuracy of FedDyn on Non-IID FashionMNIST dataset under different communication ratio

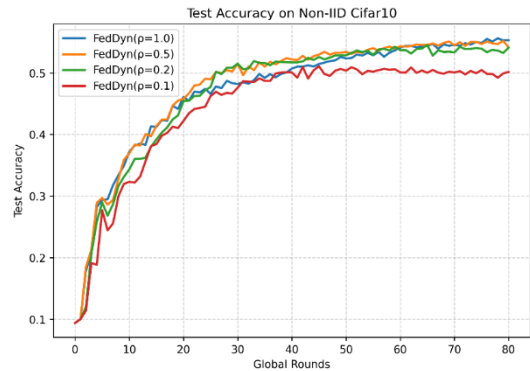


Fig. 7. The test accuracy of FedDyn on Non-IID Cifar10 dataset under different communication ratio

C. The Analysis of Communication Efficiency

To quantitatively evaluate the improvement effect of the suggested method, this paper uses the total communication cost as a metric. The total communication cost (C) is mainly determined by the amount uploaded by the client and the number of communication rounds. The formula (9) for defining the total communication cost is as follows:

$$C = R(\rho) \times \rho \quad (9)$$

where C represents the total communication cost, $R(\rho)$ represents the minimum round that the accuracy reaches 90%, 65%, 50% for MNIST, FashionMNIST and Cifar10 respectively, ρ represents single-round communication amount. Tables III, IV and V show the model performance and communication overhead of FedDyn on the Non-IID MNIST, FashionMNIST and Cifar10 datasets under different communication ratios.

The experimental results display that as the communication ratio ρ decreases, the total communication cost of system exhibits a significant downward trend. When the communication ratio is within a reasonable range, the final model accuracy of the compressed FedDyn only decreases slightly, indicating that not all the model parameters have the same importance to global model update.

TABLE III. THE MODEL PERFORMANCE AND COMMUNICATION COST OF FEDDYN ON NON-IID MNIST DATASET UNDER DIFFERENT COMMUNICATION RATIO

ρ	Best Acc	Final Acc	R@90%	Single-round amount ρ	The total cost C
1.0	0.974	0.974	10	1.0	10.0
0.5	0.967	0.952	15	0.5	7.5
0.2	0.937	0.913	18	0.2	3.6
0.1	0.946	0.925	25	0.1	2.5

TABLE IV. THE MODEL PERFORMANCE AND COMMUNICATION COST OF FEDDYN ON NON-IID FASHIONMNIST DATASET UNDER DIFFERENT COMMUNICATION RATIO

ρ	Best Acc	Final Acc	R@65%	Single-round amount ρ	The total cost C
1.0	0.72	0.71	8	1.0	8.0
0.5	0.72	0.71	10	0.5	5.0
0.2	0.70	0.69	13	0.2	2.6
0.1	0.68	0.66	16	0.1	1.6

TABLE V. THE MODEL PERFORMANCE AND COMMUNICATION COST OF FEDDYN ON NON-IID CIFAR10 DATASET UNDER DIFFERENT COMMUNICATION RATIO

ρ	Best Acc	Final Acc	R@50%	Single-round amount ρ	The total cost C
1.0	0.55	0.55	30	1.0	30.0
0.5	0.55	0.54	27	0.5	13.5
0.2	0.54	0.54	29	0.2	5.8
0.1	0.51	0.50	35	0.1	3.5

The experimental results above show that, within the framework of FedDyn, the importance of different model parameters varies significantly in the global optimization process. By introducing FedDyn-aware communication compression strategy, which only communicates key parameters, this method can achieve the effective control of communication cost without destroying FedDyn dynamic regularization structure, and maintain stable convergence and high model accuracy in Non-IID scenarios.

V. CONCLUSION

This paper addresses the issue of high communication cost in the FedDyn under the Non-IID scenarios and proposes a FedDyn-aware model compression method that is communication-effective. This method introduces parameter importance evaluation strategy without changing the original dynamic regularization optimization mechanism of FedDyn, only communicating the parameters that are more important to global model update, thereby effectively reducing the overall communication cost of the system. The experimental results on the MNIST, FashionMNIST, and Cifar10 datasets show that the proposed method can maintain model accuracy close to that of the original FedDyn while significantly reducing communication cost, thus validating the good trade-off between model performance and communication efficiency.

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